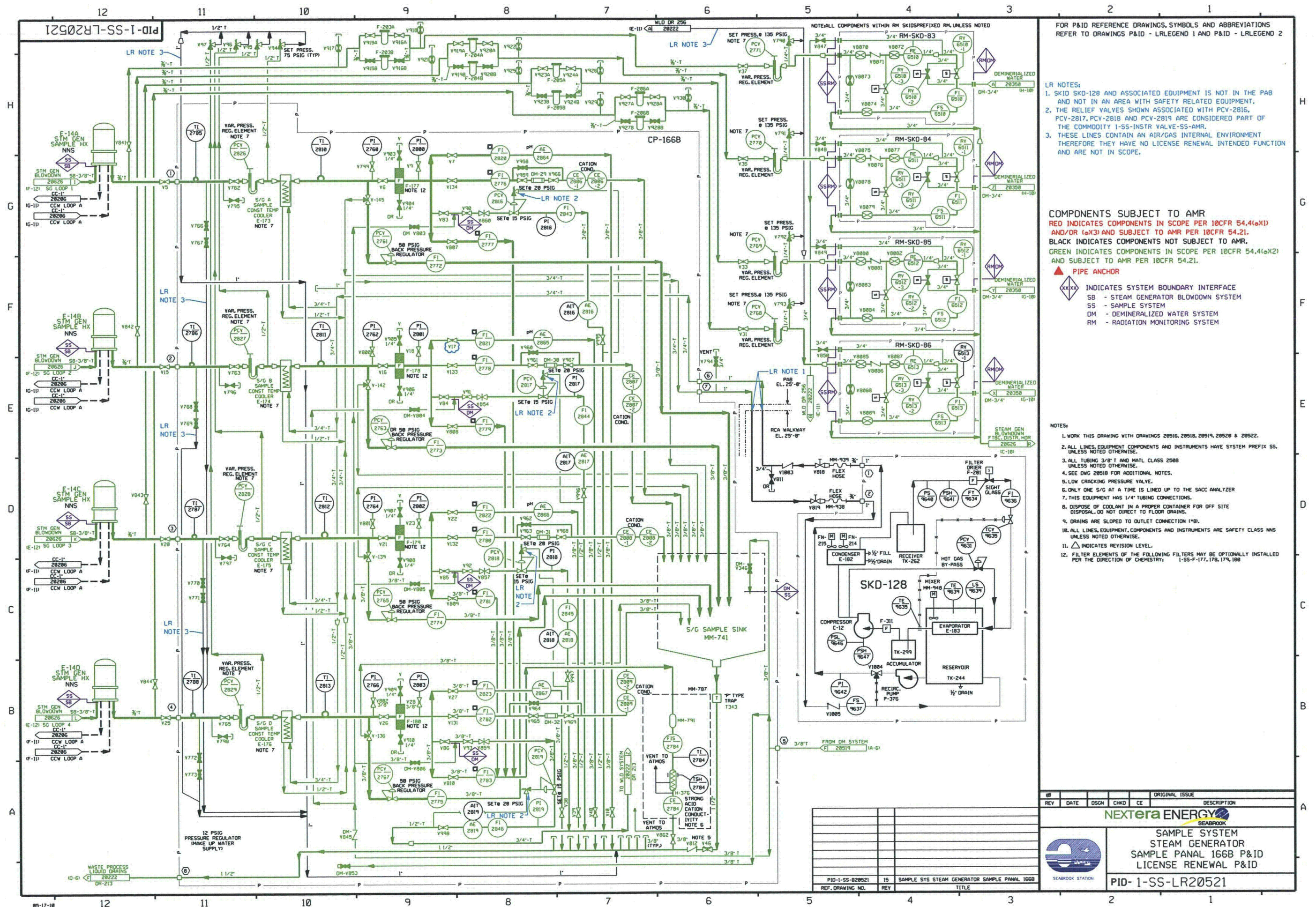
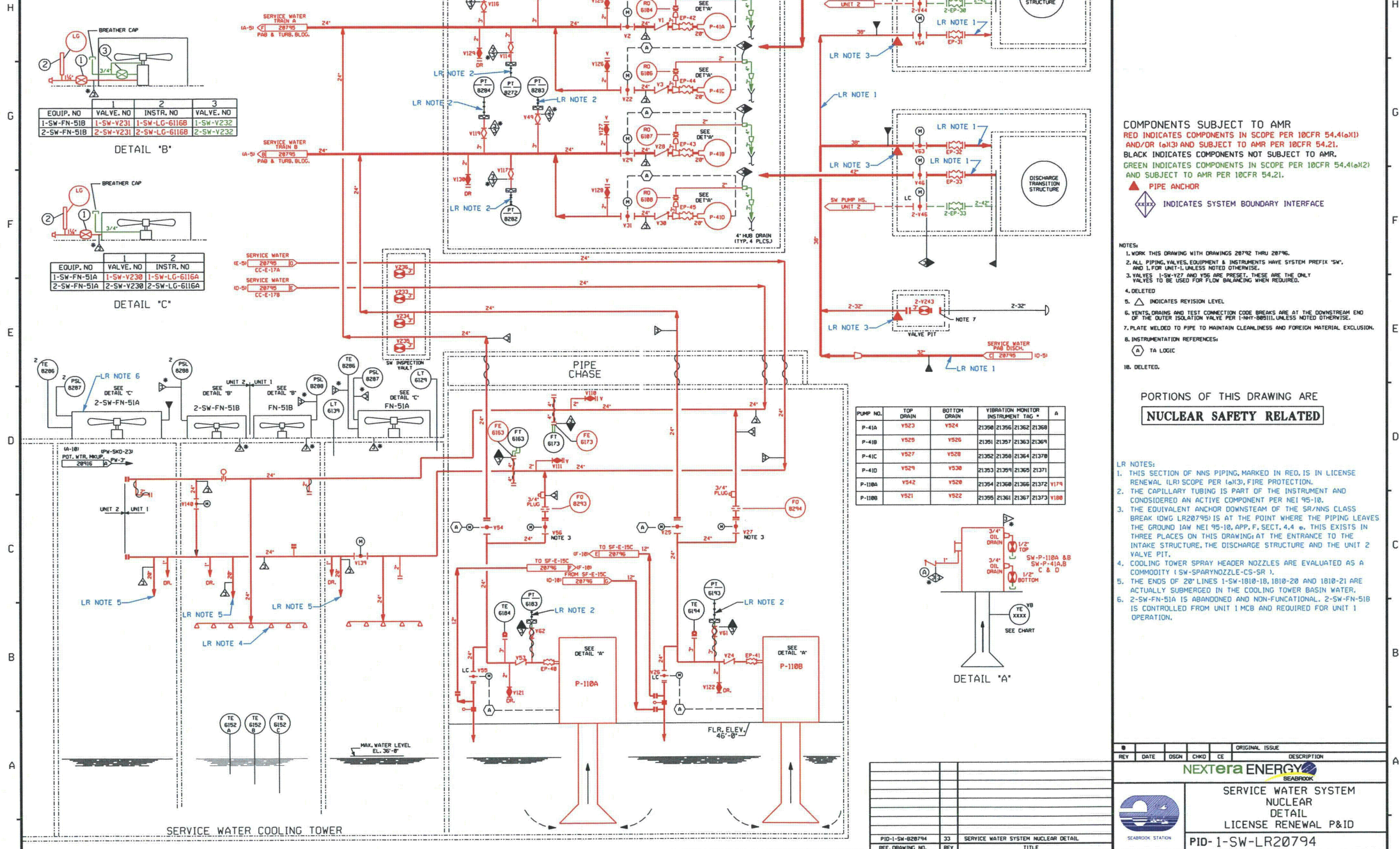




PID-1-SW-LR20794



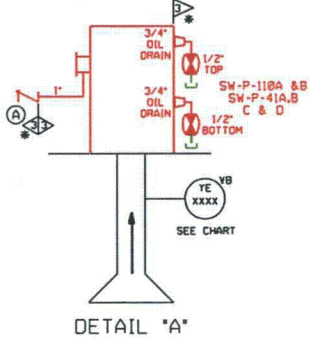
DETAIL "B"

EQUIP. NO	VALVE NO	INSTR. NO	VALVE NO
1-SW-FN-51B	1-SW-V231	1-SW-LG-6116B	1-SW-V232
2-SW-FN-51B	2-SW-V231	2-SW-LG-6116B	2-SW-V232

DETAIL "C"

EQUIP. NO	VALVE NO	INSTR. NO	VALVE NO
1-SW-FN-51A	1-SW-V230	1-SW-LG-6116A	1-SW-V231
2-SW-FN-51A	2-SW-V230	2-SW-LG-6116A	2-SW-V231

PUMP NO.	TOP DRAIN	BOTTOM DRAIN	VIBRATION MONITOR INSTRUMENT TAG #	A
P-41A	Y523	Y524	21350 21356 21362 21368	
P-41B	Y525	Y526	21351 21357 21363 21369	
P-41C	Y527	Y528	21352 21358 21364 21370	
P-41D	Y529	Y530	21353 21359 21365 21371	
P-110A	Y542	Y528	21354 21360 21366 21372	Y179
P-110B	Y521	Y522	21355 21361 21367 21373	Y180



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGENO 1 AND P&ID - LRLEGENO 2

COMPONENTS SUBJECT TO AMR
 RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
 BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
 GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.
 ▲ PIPE ANCHOR
 XXXX INDICATES SYSTEM BOUNDARY INTERFACE

- NOTES:**
1. WORK THIS DRAWING WITH DRAWINGS 20792 THRU 20796.
 2. ALL PIPING, VALVES, EQUIPMENT & INSTRUMENTS HAVE SYSTEM PREFIX "SW".
 3. VALVES 1-SW-V27 AND V56 ARE PRESET. THESE ARE THE ONLY VALVES TO BE USED FOR FLOW BALANCING WHEN REQUIRED.
 4. DELETED
 5. △ INDICATES REVISION LEVEL
 6. VENTS, DRAINS AND TEST CONNECTION CODE BREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-WHY-805111, UNLESS NOTED OTHERWISE.
 7. PLATE WELDED TO PIPE TO MAINTAIN CLEANLINESS AND FOREIGN MATERIAL EXCLUSION.
 8. INSTRUMENTATION REFERENCES:
 (A) TA LOGIC
 10. DELETED.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

- LR NOTES:**
1. THIS SECTION OF NNS PIPING, MARKED IN RED, IS IN LICENSE RENEWAL (LR) SCOPE PER (a)(3), FIRE PROTECTION.
 2. THE CAPILLARY TUBING IS PART OF THE INSTRUMENT AND CONSIDERED AN ACTIVE COMPONENT PER NEI 95-10.
 3. THE EQUIVALENT ANCHOR DOWNSTREAM OF THE SR/NNS CLASS BREAK IDWG LR20795 IS AT THE POINT WHERE THE PIPING LEAVES THE GROUND (AW NEI 95-10, APP. F, SECT. 4.4). THIS EXISTS IN THREE PLACES ON THIS DRAWING: AT THE ENTRANCE TO THE INTAKE STRUCTURE, THE DISCHARGE STRUCTURE AND THE UNIT 2 VALVE PIT.
 4. COOLING TOWER SPRAY HEADER NOZZLES ARE EVALUATED AS A COMMODITY (SW-SPARYNOZZLE-CS-SR).
 5. THE ENDS OF 20" LINES 1-SW-1810-18, 1810-20 AND 1810-21 ARE ACTUALLY SUBMERGED IN THE COOLING TOWER BASIN WATER.
 6. 2-SW-FN-51A IS ABANDONED AND NON-FUNCTIONAL. 2-SW-FN-51B IS CONTROLLED FROM UNIT 1 MCB AND REQUIRED FOR UNIT 1 OPERATION.

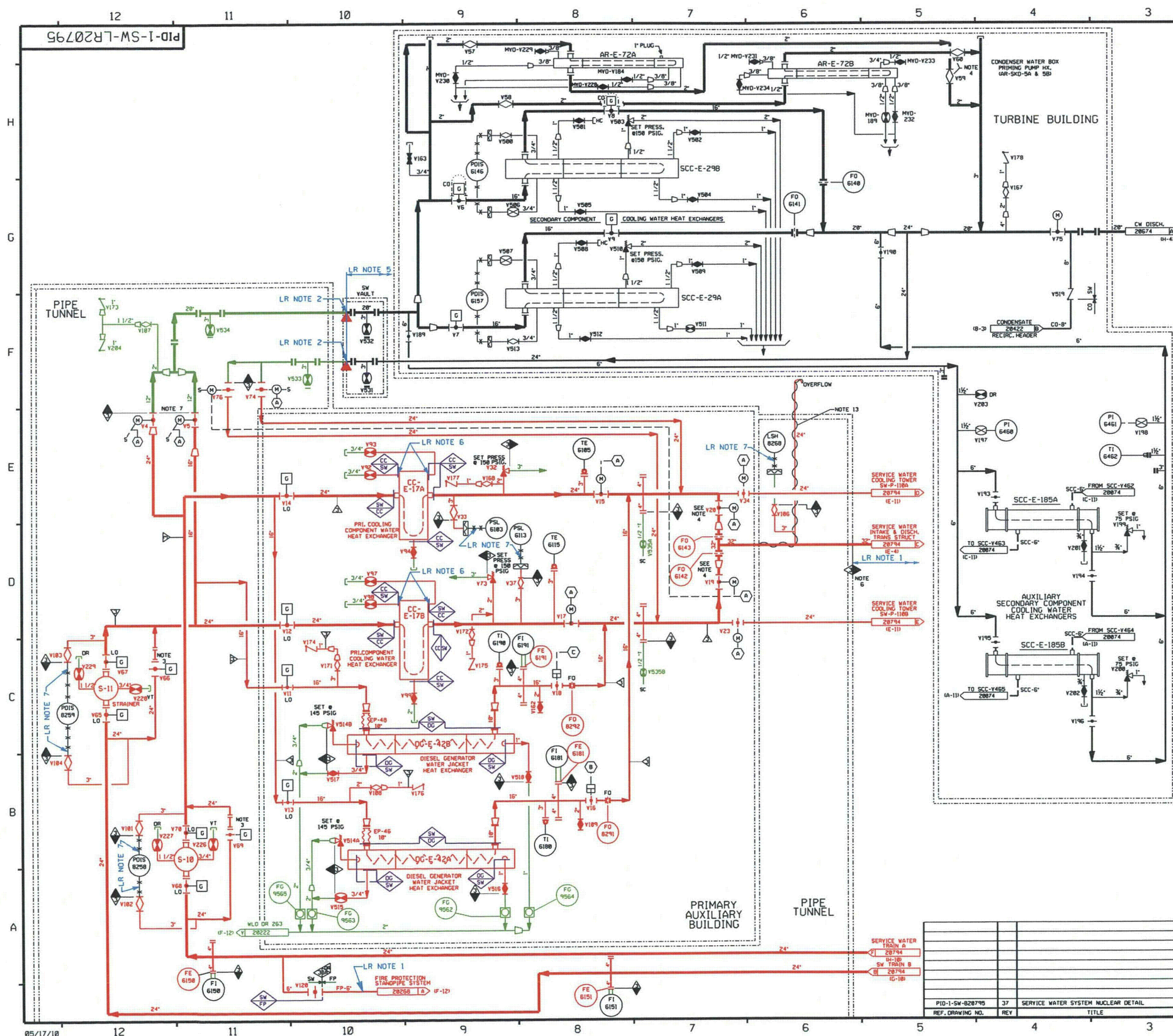
REV	DATE	DSGN	CHKD	CE	DESCRIPTION
1	05/17/10				

ORIGINAL ISSUE

NEXTERA ENERGY
 SEABROOK

SERVICE WATER SYSTEM
 NUCLEAR
 DETAIL
 LICENSE RENEWAL P&ID

PID-1-SW-LR20794



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:

- THIS SECTION OF NNS PIPING, MARKED IN RED, IS IN LICENSE RENEWAL (LR) SCOPE PER (a)(3), FIRE PROTECTION.
- THE PIPE SUPPORT EQUIVALENT ANCHOR ON THESE LINES IS THE POINT WHERE THE PIPE EXITS THE SW VAULT INTO THE GROUND (IAW NEI 95-10, APP. F, SECTION 4.4.e).
- NOT USED
- NOT USED
- NNS COMPONENTS DOWNSTREAM OF THE EQUIVALENT ANCHORS DESCRIBED IN NOTES 2, AND OUTSIDE OF THE P&B HAVE NO LR INTENDED FUNCTION AND ARE NOT IN SCOPE.
- THE SW/CC INTERFACE AT THE HEAT EXCHANGER VENT VALVES IS AT THE PIPE MATERIAL CHANGE FROM TITANIUM (CC SIDE) TO CARBON STEEL (SW SIDE).
- THE CAPILLARY TUBING IS PART OF THE INSTRUMENT AND IS CONSIDERED AN ACTIVE COMPONENT PER NEI 95-10.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.

BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.

GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR

◆ INDICATES SYSTEM BOUNDARY INTERFACE

SW - SERVICE WATER SYSTEM
CC - PRIMARY COMPONENT COOLING SYSTEM
DG - DIESEL GENERATORS
FP - FIRE PROTECTION SYSTEM

NOTES:

- WORK THIS DRAWING WITH DRAWINGS 20792 & 20794.
- ALL PIPING, VALVES, EQUIPMENT, & INSTRUMENTS HAVE SYSTEM PREFIX 'SW' UNLESS NOTED OTHERWISE.
- SPOOL PIECE NECESSARY FOR ADDITIONAL SERVICE WATER STRAINER AT LATER DATE (IF REQUIRED).
- VALVES 1-SW-V19, 20, 59 AND V60 ARE PRESET. THESE ARE THE ONLY VALVES TO BE USED FOR FLOW BALANCING, WHEN REQUIRED.
- CLASS BREAK OCCURS WHERE PIPE ENTERS EARTH.
- VALVES Y4 & Y5 TO CLOSE ON COMPLETE LOSS OF OFF SITE POWER.
- XX - DENOTES TITANIUM MATERIAL.
- DELETED
- △ INDICATES REVISION LEVEL
- INSTRUMENTATION REFERENCES
(A) TA LOGIC
(B) OPEN CMD FROM DG-1A
(C) OPEN CMD FROM DG-1B
- VENTS, DRAINS AND TEST CONNECTION CODE BREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-MHY-005111, UNLESS NOTED OTHERWISE.
- HEAT TRACING HAS BEEN DISCONNECTED AND IS NON-FUNCTIONAL.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

REV	DATE	OSGN	CHNG	CE	ORIGINAL ISSUE	DESCRIPTION
1	05/17/10					
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

PID-1-SW-20795	37	SERVICE WATER SYSTEM NUCLEAR DETAIL
REF. DRAWING NO.	REV	TITLE

05/17/10	12	11	10	9	8	7	6	5	4	3	2	1
----------	----	----	----	---	---	---	---	---	---	---	---	---

05/17/10	12	11	10	9	8	7	6	5	4	3	2	1
----------	----	----	----	---	---	---	---	---	---	---	---	---

